ATLASPDF

Universal PDF Conversion Platform

Software Requirements Specification (SRS)

A format-aware, privacy-first system for converting supported files and structured data into reliable PDF documents.

Document field	Value
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IMPORTANT SCOPE NOTE

No software can faithfully convert every possible file format into a meaningful PDF. This specification defines an explicit, testable support catalogue, safe handling for unsupported files, and phased expansion for specialised formats.

Document Control

Item	Details
Purpose	Define the functional and non-functional requirements for a multi-format file/data-to-PDF product.
Audience	Product owner, software architects, developers, QA, security/privacy reviewers, support and go-to-market team.
Status	Draft baseline for estimation and design review. Requirements remain subject to validation with users and a format-by-format proof of concept.
Change policy	Changes to supported formats, retention promises, conversion engines, security controls or service tiers require an impact review and versioned change log.
Research method	Desk research of selected global products using their public official pages and documentation. No exhaustive census of every converter and no controlled performance benchmark was performed.

Revision History

Version	Date	Change	Owner
0.1	29 Oct 2026	Initial market scan and product requirements draft.	Product / Engineering
1.0	29 Oct 2026	Baseline SRS prepared for stakeholder review.	Product / Engineering

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1. Introduction

Purpose, scope and terms

1.1 Purpose

This SRS defines what AtlasPDF must do, how it should behave, how quality will be measured and what engineering must verify before release. It is intended to be a shared contract for product, design, engineering, QA, security and operations.

1.2 Product Summary

AtlasPDF is a browser-first conversion service that turns supported documents, spreadsheets, presentations, images, web pages and structured text/data into readable PDFs. It identifies likely file types, offers relevant output settings, runs a format-specific conversion pipeline, validates the result and provides a short-lived download.

1.3 Scope Boundary

Core product promise

Convert supported inputs into a useful PDF with predictable layout, clear status, practical error messages and transparent file retention. Unknown or unsupported files must be declined safely rather than falsely advertised as convertible.

1.4 Language and Requirement Terms

Term	Meaning
MUST / SHALL	Mandatory requirement for the stated release unless explicitly marked Future.
SHOULD	Recommended requirement; deviations must be documented and justified.
MAY	Optional capability.
P1 / MVP	Required for the first public release.
P2 / Expansion	Planned after the MVP is stable and demand is validated.
P3 / Specialist	Requires format-specific engines, licensing or a separate feasibility gate.
Conversion engine	The isolated parser/rendering tool responsible for a class of formats.
Job	A conversion request and its associated state, inputs, configuration and output artifacts.
Structured data	CSV, JSON, XML, YAML and similar text formats rendered as legible tables or formatted text.

1.5 References and Research Date

Public product pages and vendor documentation were reviewed on 29 October 2026 where available. Findings represent publicly documented positioning, not an independent security audit, exhaustive feature audit or laboratory test. Full source list appears in Appendix B.

2. Product Vision, Goals and Scope

A unified conversion workflow, with honest format boundaries

2.1 Vision

Make PDF creation from everyday work files and useful data formats feel like one reliable workflow, without forcing users to learn which separate converter they need.

2.2 Product Principles

- Format-aware by design: detect the actual file type and route to the correct renderer; do not depend on the filename extension alone.
- Honest support: show supported extensions and known limitations before conversion. Never promise universal conversion for arbitrary binary or proprietary data.
- Privacy by default: no permanent file library in the MVP, short retention, visible deletion controls and minimal operational logging.
- Fidelity where practical: preserve layout, fonts, tables, page order, image orientation and spreadsheet print settings as far as the selected engine allows.
- Progressive disclosure: basic conversion should take one or two decisions; advanced settings appear only when relevant.
- Graceful failure: explain the reason, suggest a fix and preserve retry options when safe.

2.3 Business Goals

Goal ID	Goal	Measure / initial target
BG-01	Make common files convert with little effort.	At least 90% task completion in moderated usability tests for common DOCX, XLSX, PPTX and image workflows.
BG-02	Build user trust around files.	Retention policy visible before upload; 100% of completed jobs expire by the published deadline in automated checks.
BG-03	Handle common formats accurately.	At least 98% successful conversion in the approved regression corpus for supported, valid test files; quality reviewed separately.
BG-04	Keep failures understandable.	Every terminal failure maps to a documented error category and user-safe message.
BG-05	Support a sustainable service.	Per-engine cost and CPU time measured; configurable quotas and plan limits available before paid growth.

2.4 In Scope for MVP (P1)

- Responsive web application/PWA with guest conversion and optional account sign-in.
- Single-file and batch upload, drag-and-drop, file-type detection, support matrix and conversion options.
- Office documents, spreadsheets and presentations; common raster images; PDF creation from HTML and carefully sandboxed public web pages; plain text, Markdown, CSV, JSON, XML and source-code listings.
- ZIP archive intake for a bounded set of supported files, with file selection and predictable output ordering.
- Progress, cancellation where safe, retry, output preview, download and clear failure messages.
- Optional OCR for scanned images and image-only documents, initially with a small, tested language set.
- Secure temporary storage, malware scanning where feasible, isolated workers, lifecycle deletion and operational metrics.

2.5 Out of Scope for MVP

- Faithful conversion of every proprietary design, CAD, 3D, engineering or scientific binary format.
- Playback-preserving audio/video conversion into a static PDF. Future media features may create metadata sheets, contact sheets or transcript PDFs only as explicitly selected outputs.
- Conversion of an arbitrary executable, encrypted container or unknown binary into a meaningful document.
- Full PDF editor, e-signature, PDF-to-Office reconstruction, permanent cloud drive, team workspace and AI chat with files.
- Guaranteeing identical results to Microsoft Office for every complex document before the supported corpus proves it.
- Claiming legal, privacy or security certifications before a formal scope-specific audit.

3. Market Research and Competitive Analysis

Desk research across representative global products

3.1 Method and Limitations

The market contains PDF suites, general file-conversion platforms, native office exporters and offline utilities. This review compares representative products with broad international visibility based on official pages. Public claims can change, vary by region or plan and do not establish equal output quality. We did not test every product, every plan or every supported conversion pair.

3.2 Competitive Snapshot

Product / category	Publicly documented scope	Strengths / observed positioning	Product lesson
Adobe Acrobat Online / Desktop	Common Office, image and document formats; desktop Acrobat has a wider format table and some dependencies on installed apps. Online converter offers major everyday types. [R01][R02]	Known brand, mature PDF workflow, emphasis on retaining layout; limitations are linked to format, app, font and feature support.	List supported formats by route and explain when advanced formatting may change.
Smallpdf	PDF suite with Office, image, HTML, text, OpenDocument, EPUB and other listed tools; OCR and PDF/A features are also listed. [R03][R04]	Easy conversion journey and adjacent PDF tools. Its public converter describes 20+ formats; scope depends on chosen tool.	Keep the default flow short, then offer useful PDF actions after conversion without making them prerequisites.
iLovePDF	Office-to-PDF plus a broad set of PDF actions; web, mobile and desktop are included in its paid offering. Basic tier has limited processing and tool quotas. [R05]	A freemium multi-device suite with clear plan limits.	Design quotas as explicit product configuration, not hard-coded scattered limits.
CloudConvert	File conversion API advertises 200+ formats, advanced options, async jobs, webhooks and storage integrations. [R06]	Useful for format diversity, integration and custom workflows. Its privacy page states automatic deletion no later than 24 hours for uploaded jobs. [R07]	Use an engine-adapter architecture, job queue, webhooks and configurable format-specific options. Aim for shorter published retention.
Convertio	Advertises 300+ file formats and 25,600+ conversion pairs; its policy describes input/temp deletion after conversion and output retention up to 24 hours unless deleted sooner. [R09][R10]	Broad web-based conversion with custom options for many conversion types.	Show a live support catalogue and state the deletion window in plain language.
Zamzar	Advertises 1,100+/1,200+ formats across document, image, video, audio, ebook, archive and CAD categories; offers a conversion API. [R11][R12]	Large catalogue and API reach, including specialist file categories.	Separate “render the file as a PDF” from “extract metadata/frames/transcript as a PDF.” Do not imply every source type has an equivalent PDF representation.
PDF24 Creator / Tools	Free Windows offline app keeps files on the user PC. Its online webpage-to-PDF converter processes pages in the cloud and deletes outputs after a short period. [R13] [R14]	Offline processing is a strong privacy option; web tools offer a quick no-install route.	Treat offline/local processing as a credible future differentiator; disclose cloud vs local processing per route.
FreeConvert	General file conversion; its privacy page states files are stored on AWS in Ireland and automatically deleted after 8 hours, with manual deletion available sooner. [R15]	Clear retention policy and broad conversion categories.	Make deletion controls easy to find and verify them with automated lifecycle tests.

Product / category	Publicly documented scope	Strengths / observed positioning	Product lesson
Microsoft Office / LibreOffice	Office apps and LibreOffice provide native PDF export options. Microsoft documents that spreadsheet internal links can be lost and that font substitution can occur in some online scenarios; LibreOffice exposes export options. [R16][R17]	Often the best starting point for layout fidelity in files created in the same office suite; works locally.	Benchmark against native reference output and expose print area, page fit, font and export warnings.

3.3 What the Research Suggests

- Format count alone is an incomplete quality metric. A platform may support hundreds of formats while complex layouts or specialised files still require a particular engine or application.
- Competitors cluster into three broad models: polished PDF utilities, large format APIs and local/native export tools. AtlasPDF can unify the first two for everyday formats and add local-first processing later.
- Privacy practices differ: published deletion windows in the reviewed examples range from immediate input deletion to automatic cleanup within 8 or 24 hours. AtlasPDF should make its own retention promise specific and testable.
- Users need to know what will happen before they upload: a scanned PDF may require OCR, a GIF may become still images or a selected frame sequence, and an audio/video file cannot be represented faithfully as static pages without choosing an extraction workflow.
- Conversion controls should adapt to the source type. Spreadsheet print-area and fit options matter for XLSX; image order and orientation matter for image sets; page-breaking and resource loading matter for HTML.

3.4 Suggested Positioning

Positioning statement

“One clear route from common files and useful data to a clean PDF, with visible support limits, quality checks and short-lived file storage.” The differentiator should be trust and workflow clarity, backed by measurable format quality, rather than an unqualified “supports everything” claim.

3.5 Feature Opportunity Matrix

Opportunity	User value	Delivery
Pre-upload support check	Reduces failed uploads and surprises.	P1: extension + MIME/signature detection and support catalogue.
Format-aware options	Improves output without overwhelming users.	P1: show only relevant controls.
Quality warnings	Helps users catch layout or content loss.	P1: per-engine warnings and preview; P2: more automated comparisons.
Short retention + delete button	Builds confidence with sensitive documents.	P1: object lifecycle + user deletion + automated verification.
Offline / local conversion	Useful for confidential files and low-bandwidth users.	P2: desktop/local runtime; limited browser-local conversions may be explored.
Developer API and webhooks	Allows teams to automate routine conversion.	P2/P3 after core quality and operations are stable.
Specialist formats	Serves niche professional workflows.	P3, each format subject to business demand, licensing and a proof of concept.
Freemium tiers	Balances useful free access with sustainable processing cost.	Start with configurable caps; test paid options for larger batches, OCR, priority, API and enterprise controls. Set prices after cost and user research.

4. Users, Personas and Operating Environment

Who the product serves and where it runs

4.1 User Classes

Persona	Primary jobs	Needs / constraints
Student / applicant	Convert assignments, scanned notes, slides and images.	Fast, low-cost, works on mobile, sensible page layout and OCR.
Office user	Convert reports, spreadsheets, forms and presentations.	Formatting, sheet selection, print area, consistent page numbering and confidentiality.
Developer / analyst	Turn Markdown, source code, CSV, JSON, XML or logs into readable PDF evidence.	Monospace rendering, line wrap, page headers, long-file handling and deterministic output.
Privacy-conscious user	Convert a contract, ID scan or client file.	Clear retention/deletion, no public links, minimal logs and no AI processing by default.
Small-team administrator	Set limits, inspect failure rates and manage billing/API later.	Usage visibility, stable queues, cost controls and access separation.

4.2 Operating Environment

- Client: modern desktop and mobile browsers, responsive web experience; PWA installability is desirable but not required to use the service.
- Initial browser support: latest two stable versions of Chrome, Edge, Firefox and Safari, plus current iOS Safari and Android Chrome, subject to test capacity.
- Server: containerized, Linux-based conversion workers with constrained CPU, memory, disk, process count and wall-clock execution time.
- Network: HTTPS only for application and download traffic; object storage is private; time-limited signed download URLs are used for outputs.
- Locale: English at MVP, with UTF-8 end-to-end and architecture ready for right-to-left text and language-specific OCR expansion.

4.3 Assumptions and Dependencies

- Users have legal permission to process the files they submit.
- An explicit format catalogue is maintained with extension, signature, parser, engine version, limitations, maximum size, timeout and test corpus.
- Conversion engines and their dependencies can be redistributed and operated under compatible licenses; legal review is required before production.
- Network access and cloud-region choice will be reviewed for privacy, latency, cost and applicable legal obligations.
- Complex Office output fidelity must be verified with representative test files from real users; file extension coverage does not prove fidelity.

5. Use Cases and User Journeys

Primary flows and alternate paths

5.1 UC-01: Convert a Single Document

1. User opens the converter and chooses a file or drags it into the upload area.
2. Client and server inspect size, extension, MIME hints and file signature. The system identifies a supported type or explains that it is not supported.
3. System presents source-aware options such as page size or spreadsheet print settings where applicable.
4. User starts conversion. A job ID is created and the UI displays a truthful state: uploading, queued, processing, validating, ready or failed.
5. The worker creates a PDF in an isolated environment, validates it and stores the result temporarily.
6. User previews, downloads, deletes or retries. The system expires input and output according to the published retention deadline.

5.2 UC-02: Batch Conversion

The user selects multiple files. The system reports which files are supported, allows per-file settings where needed, and offers either one PDF per input or a single merged PDF only when all selected inputs can be rendered. Output order follows the visible queue order. A single file failure must not silently corrupt successful outputs.

5.3 UC-03: Convert a ZIP Archive

The user uploads a ZIP. The system lists entries without extracting to a public path, calculates archive and expanded-size limits, rejects path traversal and excessive nesting, and lets the user select supported entries. The system does not execute archive contents. A configured policy prevents archive bombs and recursive extraction loops.

5.4 UC-04: Convert a Public Web Page

The user pastes a URL. The system validates the URL, blocks internal/private destinations and unsafe redirects, applies DNS/IP checks at connection time, restricts protocols and resource limits, and renders only within a network-isolated browser worker. Failures such as authentication walls, blocked resources or very long pages are explained. This feature is disabled until SSRF controls pass security review.

5.5 UC-05: Create a Searchable PDF from a Scan

The user uploads an image or image-only document, selects OCR and an available language, then converts. The result is marked as OCR-processed. The UI warns that recognition may be imperfect, especially for low-resolution, handwritten or skewed content. Source images remain available only until expiry.

5.6 UC-06: Unsupported or Failed File

The system names the reason category without exposing stack traces: unsupported format, damaged file, password-protected/encrypted input, detected malware, resource limit, timeout or engine failure. It suggests a safe next action and supplies a support code that does not expose file content.

6. Functional Requirements

Normative, testable requirements

Priorities: P1 = MVP; P2 = expansion; P3 = specialist capability. Every requirement below must map to at least one test case or an explicit deferral before release.

6.1 File Intake and Identification

ID	Priority	Requirement	Verification
FR-001	P1	The system SHALL accept files through file picker and drag-and-drop on supported browsers.	UI test on desktop and mobile.
FR-002	P1	The system SHALL allow a configurable number of files per batch and display the batch limit before upload.	Limit displayed and enforced server-side.
FR-003	P1	The system SHALL enforce per-file, per-batch and expanded-archive size limits server-side.	Oversized and ZIP-bomb fixtures rejected.
FR-004	P1	The system SHALL inspect file signature/content in addition to filename extension and client-supplied MIME.	Renamed/mismatched fixtures detected.
FR-005	P1	The system SHALL maintain a support catalogue that maps formats to engines, options, limits and known caveats.	Catalogue powers UI and dispatch.
FR-006	P1	The system SHALL identify unsupported, corrupted, empty and password-protected files before or during safe parsing, then show actionable errors.	Each condition has a stable error code.
FR-007	P1	The system SHALL replace user filenames with generated storage identifiers and safely encode the display name.	No path traversal or overwrite from filename.
FR-008	P1	The system SHALL allow users to remove a file from the queue before conversion begins.	Removed item is not processed.
FR-009	P2	The system SHOULD support importing from approved cloud providers after OAuth scopes and security review.	Tokens least-privilege; import is explicit.

6.2 Conversion and Job Lifecycle

ID	Priority	Requirement
FR-010	P1	The system SHALL create a unique job ID and track queued, processing, validating, ready, failed, cancelled and expired states.
FR-011	P1	The system SHALL dispatch work by detected format and selected conversion profile, not by extension alone.
FR-012	P1	The system SHALL expose progress at meaningful stages; it SHALL NOT invent a precise percentage when the engine cannot provide one.
FR-013	P1	The system SHALL allow cancellation when safe. A cancellation request SHALL stop worker execution or terminate the isolated job within a configured deadline.
FR-014	P1	The system SHALL retry only transient failures using bounded backoff and idempotent job keys; deterministic parse failures SHALL NOT be retried indefinitely.
FR-015	P1	The system SHALL support one PDF per input and a merged PDF option for compatible batches, with user-visible ordering.
FR-016	P1	The system SHALL generate a unique output artifact per job and validate that it is a readable PDF with at least one page unless an approved exception applies.
FR-017	P1	The system SHALL permit download only to the job owner/authorized session using short-lived, unguessable or signed access.
FR-018	P1	The system SHALL provide a preview and download action after success, without requiring the user to create an account for basic guest conversion.

ID	Priority	Requirement
FR-019	P1	The system SHALL provide a safe retry action that reuses valid settings and never duplicates billing/usage consumption for a single accepted job unless policy says otherwise.
FR-020	P2	The system SHOULD support webhooks and API-based job creation after authentication, rate limits and tenant isolation are implemented.

6.3 PDF Output Controls

ID	Priority	Requirement
FR-021	P1	The system SHALL provide a sensible default profile and preserve the source layout where supported by the selected engine.
FR-022	P1	The system SHALL expose page size (A4, Letter, source/auto where available), orientation and margins when the engine supports these options.
FR-023	P1	The system SHALL provide source-specific settings for spreadsheet sheets/print area/scaling and image ordering/orientation when relevant.
FR-024	P1	The system SHALL allow users to select image fit mode (contain/cover or fit-to-page) and image resolution preset for raster-image conversion.
FR-025	P1	The system SHALL support common page-range or slide-range controls when supported by the source format and engine.
FR-026	P1	The system SHALL preserve Unicode text using embedded or compatible fonts where licensing and engine support permit; substitutions SHALL be disclosed when detectable.
FR-027	P1	The system SHALL provide optional OCR for supported image inputs and scanned documents, with language selection and a clear OCR label.
FR-028	P1	The system SHALL apply safe defaults for animated or multi-frame image inputs and explain whether frames are preserved as separate pages or a selected frame is used.
FR-029	P2	The system SHOULD offer PDF/A output profiles after conformance testing and explain that archival profiles may disable certain features or change file size.
FR-030	P2	The system SHOULD support accessible PDF output improvements, including document title/language and semantic tagging when the source and engine support them.

6.4 Format-Specific Capabilities

ID	Priority	Requirement
FR-031	P1	Office documents and presentations SHALL render through a controlled document engine with explicit page and font handling.
FR-032	P1	Spreadsheets SHALL support sheet selection and a fit/scaling choice; the UI SHALL warn when tables may become too small or span many pages.
FR-033	P1	CSV, JSON, XML and plain text SHALL be rendered as readable content, with line wrapping, page headers where selected and a readable overflow strategy.
FR-034	P1	Markdown and supported source-code files SHALL be rendered in a document layout; source-code mode SHALL use a legible monospaced font and preserve line order.
FR-035	P1	HTML uploads SHALL render with local resources only by default; remote resources require a controlled policy and must not access internal networks.
FR-036	P1	Public URL to PDF SHALL remain feature-flagged until URL parsing, DNS/IP enforcement, redirect handling, protocol restrictions and sandbox egress tests pass.
FR-037	P1	ZIP conversion SHALL list supported entries, preserve a deterministic sort order, reject unsafe paths and enforce expanded size/depth/file-count limits.
FR-038	P2	The system MAY add EPUB, email formats, vector formats and additional image types after per-format conformance testing.
FR-039	P3	Audio/video sources MAY produce a transcript, metadata sheet or selected-frame contact sheet only as a distinct extraction workflow, not as an equivalent static conversion.

ID	Priority	Requirement
FR-040	P3	CAD, 3D and specialist proprietary formats SHALL require a separately approved engine and commercial/legal review before support is advertised.

6.5 Privacy, Accounts and Administration

ID	Priority	Requirement
FR-041	P1	The system SHALL support guest conversion without mandatory account creation for the advertised guest limits.
FR-042	P1	The system SHALL display file retention and processing location information before upload and offer an immediate delete action for each job.
FR-043	P1	The system SHALL delete inputs, outputs and temporary files automatically within the published retention limit, including failed jobs.
FR-044	P1	Operational logs SHALL exclude file contents, passwords, access tokens and raw signed URLs. File names SHALL be omitted or pseudonymized in logs unless justified.
FR-045	P1	The system SHALL provide configurable quotas for guest, registered and API users, including file size, batch count, daily jobs and OCR usage.
FR-046	P1	Administrators SHALL be able to enable/disable a format or engine, adjust limits and disable a vulnerable engine without a full application release.
FR-047	P1	Administrators SHALL see job outcome, duration, engine version, resource use and a safe error code, not uploaded file content.
FR-048	P2	The system SHOULD support authenticated accounts, subscriptions, invoices and usage history after privacy and billing decisions are approved.
FR-049	P2	The system SHOULD support organisation-level access controls, audit logs and regional processing options for business plans.
FR-050	P2	The system MAY support user-selected persistent workspaces only with clear retention controls and a separate consent/plan model.

6.6 Accessibility and Help

- FR-051 P1: All primary actions SHALL be keyboard-accessible and expose semantic labels, focus states and accessible status announcements.
- FR-052 P1: Error messages SHALL name the problem and a next action; they SHALL not expose internal paths, stack traces or secrets.
- FR-053 P1: The UI SHALL describe format support and limitations in plain language, including known output changes.
- FR-054 P1: The application SHALL support responsive layouts and respect reduced-motion preferences.
- FR-055 P2: The product SHOULD provide translated UI and OCR language packs based on measured user demand.

7. Supported Format Catalogue and Conversion Rules

Explicit support is part of the product contract

7.1 Proposed MVP Format Matrix

Category	Extensions / inputs	PDF behavior	Release gate
Office word processing	DOCX, DOC, RTF, ODT	Convert to paginated PDF using isolated office renderer.	P1, subject to renderer regression corpus.
Spreadsheets	XLSX, XLS, ODS, CSV	PDF with sheet selection, print area/scaling options; CSV rendered as table or fixed-width text.	P1; formulas/recalc behavior must be documented.
Presentations	PPTX, PPT, ODP	Slides as pages; select slide range. Speaker notes excluded by default.	P1.
Images	JPG/JPEG, PNG, TIFF, BMP, WebP	One or more images to pages; respect orientation metadata; configurable fit and order.	P1; verify each engine/library format.
Text and markup	TXT, MD, HTML	Readable document layout with page breaks, wrapping and optional headers.	P1.
Structured data	JSON, XML, CSV	Prettified text, collapsible structure is not required in PDF; very long lines need wrapping/overflow strategy.	P1.
Source code / logs	Common text-based source extensions, LOG	Code-listing mode with monospaced font, line numbers optional, long-line policy.	P1; extension allowlist maintained.
Web page URL	HTTP/HTTPS public pages only	Full-page capture, with best-effort external resources and warning when blocked.	P1 after security gate only.
ZIP archive	ZIP only	Select supported entries, individual PDFs or merged PDF in deterministic order.	P1 with strict resource and path controls.
OCR inputs	Raster images and supported scan PDFs	Searchable PDF with text layer and source image retained where engine supports it.	P1 with tested language packs.
EPUB / email / SVG / HEIC	Format-specific handling	Paginate book/email or render vector/image content.	P2 after quality and license review.
Audio / video / CAD / 3D / proprietary packages	No generic static PDF equivalent	Potential metadata report, contact sheet, transcript or specialised flattened output.	P3 / individually approved; not advertised in MVP.

7.2 Format Registry Fields

Each format entry SHALL include: canonical format ID; accepted extensions; signature/magic detection rules; permitted MIME hints; owning engine; engine and dependency version; supported options; file/page/expanded-size limits; expected output behavior; known limitations; applicable license; supported locales; test-fixture set; and feature flag state.

7.3 Format Admission Checklist

- A valid and representative test corpus is available, including boundary, malformed, large and multilingual files.
- Output quality is compared with a suitable reference renderer or agreed visual/content expectations.
- Engine license, redistribution, font license and transitive dependencies are reviewed.
- Parser can run with no unnecessary network access, restricted filesystem access and resource ceilings.
- Failure cases, timeout behavior, telemetry and user-facing caveats are implemented.
- Security testing covers malformed input, decompression bombs, parser crashes and cross-job isolation.

7.4 Business Rules

Rule	Definition
BR-01	An extension alone never proves that a file is safe or supported.
BR-02	Unknown binary files are never silently rendered as text; a hex dump is not a standard conversion.
BR-03	Label each route as native render, text/data presentation, OCR-assisted or extraction workflow.
BR-04	Mark output ready only after PDF validation; batch order follows the visible order.
BR-05	Report batch failures individually; keep valid outputs until their normal expiry.
BR-06	Never log passwords. Disclose and obtain consent before using third-party processing or AI.

8. Non-Functional Requirements

Measurable quality attributes

8.1 Performance and Capacity

ID	Requirement / proposed baseline	Measurement method
NFR-PERF-01	For files up to 20 MB and up to 20 pages in the approved common-office benchmark corpus, target p95 conversion completion within 30 seconds after upload at normal load. Complex spreadsheets and OCR are measured as separate classes.	Automated benchmarks by format/engine and load band.
NFR-PERF-02	API requests for metadata/status/options should target p95 under 500 ms, excluding file upload, conversion and third-party identity/billing services.	APM distributed traces.
NFR-PERF-03	Initial pilot should sustain 100 concurrent conversion jobs without cross-tenant data leakage or unbounded queue growth; capacity must be re-baselined from observed cost and load tests.	Load, soak and queue saturation tests.
NFR-PERF-04	Large uploads should support resumable or multipart upload where the storage provider and client permit it.	Network interruption test.
NFR-PERF-05	Worker CPU, memory, scratch disk and wall-clock limits SHALL be enforced per job.	Container resource and timeout tests.

8.2 Availability and Recovery

- NFR-AVL-01 P1: Target 99.5% monthly availability for the conversion control plane during public beta, excluding published maintenance and external dependencies, measured with synthetic checks.
- NFR-AVL-02 P1: Queue outage or worker failure must not lose accepted job state; jobs shall move to retryable or failed status without appearing complete.
- NFR-AVL-03 P1: Deployments must support rollback to a known-good API and worker image. Engine upgrades are versioned and can be disabled through configuration.
- NFR-AVL-04 P1: Since files are temporary, file contents are not backed up. Job metadata backup/restore policy is separate and must not reintroduce expired artifacts.
- NFR-AVL-05 P2: 99.9% availability and multi-region recovery are target candidates only after usage, cost and operational readiness justify the investment.

8.3 Security

- NFR-SEC-01: All production traffic must use HTTPS. Storage objects must remain private by default and be retrievable only through authorized, short-lived access.
- NFR-SEC-02: Validate file type through signature/content inspection; enforce file count, size, expanded-size, page count, timeout and CPU/memory limits.
- NFR-SEC-03: Parse untrusted files inside isolated containers with a non-root identity, read-only base filesystem, per-job scratch directory, limited process count and restricted/no outbound network access.

- NFR-SEC-04: Scan uploads where operationally feasible; reject confirmed malicious files. Scanner failure policy must be explicit and fail closed for high-risk formats where required.
- NFR-SEC-05: Use application-generated storage names; protect against path traversal, archive traversal, XXE, XML expansion bombs, command injection and parser vulnerabilities.
- NFR-SEC-06: URL rendering must enforce HTTP/HTTPS allow rules, DNS/IP validation, redirect revalidation, block loopback/private/link-local/metadata ranges and deny unnecessary protocols. Security gate before launch.
- NFR-SEC-07: Secrets reside in a managed secret store; they must not be placed in source, logs, job metadata or client code.
- NFR-SEC-08: Apply least privilege, CSRF protections where cookie-auth is used, rate limits, bot/abuse controls, dependency scanning and regular patching.
- NFR-SEC-09: Security events and administrative changes are auditable without recording document content.
- NFR-SEC-10: Run an independent threat model and penetration test before public launch; critical/high findings must be resolved or explicitly risk-accepted by the accountable owner.

8.4 Privacy and Retention

ID	Requirement
NFR-PRV-01	Default processing is temporary. Inputs, intermediate artifacts and outputs shall be automatically deleted within 60 minutes of job terminal state; failed jobs are covered. This is the proposed product promise and must be verified end-to-end before publication.
NFR-PRV-02	A “Delete now” action SHALL immediately revoke download access and enqueue deletion; UI SHALL confirm only after the control plane accepts the request, and final deletion status is observable internally.
NFR-PRV-03	Job artifact retention, application logs, account profile data and billing records have separate documented schedules.
NFR-PRV-04	Do not use submitted files to train AI models. OCR is local to the isolated worker unless a separately disclosed, opt-in third-party service is approved.
NFR-PRV-05	The privacy notice must describe processing region, subprocessors, file retention, metadata logs, account data and user rights in plain language.
NFR-PRV-06	Privacy/legal review must map obligations for launch markets, including relevant data protection and consumer rules; no certification or blanket legal compliance claim is implied by this SRS.

8.5 Usability and Accessibility

- NFR-UX-01: A first-time user should complete a common conversion without an account and without reading a tutorial.
- NFR-UX-02: The landing/upload screen should communicate supported formats, maximum limits, privacy policy and the next step before the user uploads.
- NFR-A11Y-01: Target WCAG 2.2 AA for the web UI, using automated checks plus keyboard and screen-reader testing. [R20]
- NFR-A11Y-02: Focus order, text contrast, visible labels, status announcements, error association and reduced-motion support must be tested.
- NFR-LOC-01: All content must use UTF-8 and layout must tolerate long filenames, Unicode scripts and translated labels.
- NFR-UX-03: Mobile layout must support file chooser access, queue management and download without hover-only controls.

8.6 Maintainability and Portability

- NFR-MNT-01: Each conversion engine must be hidden behind a stable adapter contract for detect, validate, convert, inspect output and report errors.
- NFR-MNT-02: Conversion engine versions and fonts must be pinned, scanned and promoted through regression testing before production.
- NFR-MNT-03: The application must separate product UI, job orchestration, file storage and rendering workers so one engine can be disabled or replaced independently.
- NFR-MNT-04: CI/CD must include unit, integration, contract, security and visual-regression tests; release artifacts must be traceable to a source revision.
- NFR-OBS-01: Metrics and traces must correlate API request, job and engine version using opaque IDs and must not include file content or sensitive filenames.

9. System Architecture and Technical Design

A modular conversion pipeline

9.1 Recommended Architecture

Recommended first implementation: a responsive Next.js/TypeScript web app, ASP.NET Core API, PostgreSQL for job metadata, private Azure Blob Storage for temporary artifacts, Azure Service Bus for job dispatch and Linux container workers on Azure Container Apps Jobs (or an equivalent container platform). Use a managed secret store, OpenTelemetry/Application Insights and a separate engine adapter for each conversion family. The architecture is cloud-neutral enough to migrate, but the first deployment should choose one cloud and one primary region.

Layer	Components	Responsibilities
Experience	Next.js web app / PWA	Upload, settings, live state, preview, download, privacy/help content.
API/control plane	ASP.NET Core API; identity; quota service	Auth/session, support catalogue, job create/status/cancel/delete, signed upload/download access, plan limits.
Storage	Private object storage	Input, intermediates and output under generated keys; lifecycle rules; no public buckets.
Queue/orchestration	Message queue + job state machine	Bounded concurrency, retry/backoff, cancellation, dead-letter path, idempotency.
Workers	Isolated container pool	Signature validation, malware scan, format engine execution, PDF validation, output metadata.
Engine adapters	Office, image, markup, data, OCR, URL renderer	Format-specific conversion with pinned versions, resource limits and caveat metadata.
Platform services	PostgreSQL, secrets, logs, metrics, alerts	Metadata only, secret handling, audit trail, service health and cost telemetry.

9.2 Job Pipeline

1. Create job record with opaque ID, requested mode, selected settings, quota reservation and retention deadline.
2. Issue short-lived upload permission for a private object key; verify server-side size and upload completion.
3. Inspect content and signature. Quarantine or reject unknown/suspicious inputs. Record only safe metadata.
4. Route the file to a worker image that matches the approved format registry entry.
5. Run the engine in isolation with resource ceilings and controlled scratch storage; capture progress and a safe error category.
6. Inspect output: confirm PDF signature/structure, page count, non-empty content where expected, page bounds and successful read/preview.
7. Store output privately and return a time-limited download link. Delete intermediates promptly and schedule final cleanup for all artifacts.
8. Emit metrics for duration, size, pages, engine version, result category and resource cost; exclude file contents.

9.3 Engine Strategy

Format family	Candidate implementation	Decision / caution
Office files	LibreOffice headless isolated workers for initial POC.	Test fonts, charts, formulas, macros, external links and print settings. Disable macro execution; consider a commercial engine only after fidelity/cost review.

Format family	Candidate implementation	Decision / caution
HTML / web pages	Sanitized local HTML pipeline; Chromium/Playwright for controlled URL capture.	URL capture has SSRF risk and must remain feature-flagged until security tests pass.
Images	libvips or another well-maintained image library.	Restrict formats, dimensions, memory and metadata handling; re-encode when appropriate.
Markdown / source / data	Custom HTML templates or Pandoc with strict input policy, rendered in headless browser or PDF engine.	Prevent template injection; wrap long lines and escape source content.
OCR	Tesseract + OCRmyPDF or an equivalent tested toolchain.	Language-pack licensing, recognition accuracy, time and memory vary; scanned output needs per-language test sets. [R21]
PDF assembly / validation	A vetted PDF library and command-line tool selected after license/security review.	Do not select solely on convenience; track dependencies and CVEs.

9.4 Engineering Rules

- Never run arbitrary commands assembled from user-controlled filenames or options.
- Never mount the host Docker socket or cloud credentials into untrusted conversion containers.
- Deny worker egress by default; allow only explicitly required, pinned endpoints for approved workflows.
- Use one job per scratch directory and destroy the container/volume after completion.
- Pin fonts and engine versions; any engine change requires golden-file comparison and security review.
- Use asynchronous job processing for conversion; API requests should return a job ID quickly rather than hold a request open through rendering.

10. Data Model and API Requirements

Minimal metadata, clear job ownership

10.1 Data Entities

Entity	Key fields	Rules
User / GuestSession	user_id or session_id, plan_id, created_at, status	Guest session must be opaque, rate-limited and isolated.
Job	job_id, owner_ref, input_key, output_key, input_type, output_type, engine_id, state, progress_stage, created_at, terminal_at, expires_at, error_code	No raw content, passwords or signed URLs. Input/output keys are generated.
JobFile	file_id, job_id, display_name_safe, detected_format, size_bytes, page_count, order_index, status	Safe display name is optional; avoid unnecessary filename logs.
ConversionProfile	profile_id, format_id, options_schema, defaults, max_bytes, max_pages, timeout, version	Settings validated against server-owned schema.
FormatRegistry	format_id, extensions, signature_rule, engine_id, version, status, caveats, license_review_status	Feature flag can disable a format/engine quickly.
QuotaLedger	owner_ref, period, jobs_used, bytes_used, OCR_units, reservations	Reservation is idempotent; failed jobs follow documented charge/refund policy.
AuditEvent	actor_ref, action, object_id, timestamp, request_id, outcome	No user file content or long-lived access credential.

10.2 API Surface (Proposed)

Method / path	Purpose	Controls
GET /api/v1/formats	Return active supported formats and relevant limits/options.	Cacheable; no internal engine paths.
POST /api/v1/jobs	Create job and reserve quota.	Auth/session, schema validation, idempotency key, rate limit.
POST /api/v1/jobs/{id}/upload-ticket	Issue a private short-lived upload URL or multipart session.	Owner authorization, expected size/type, expiration.
POST /api/v1/jobs/{id}/complete-upload	Finalize upload and queue validation.	Recheck object existence/size; only one transition.
GET /api/v1/jobs/{id}	Read state, stage, warnings, expiry and safe error.	Owner-only, no private storage key.
POST /api/v1/jobs/{id}/cancel	Request cancellation.	Owner-only, idempotent.
POST /api/v1/jobs/{id}/delete	Revoke access and request artifact deletion.	Owner-only; record deletion state.
GET /api/v1/jobs/{id}/download	Return a short-lived download redirect after ready-state validation.	Owner-only; no permanent public URL.
GET /api/v1/admin/metrics	Admin operational summary.	Strong authentication, role authorization and audit.

10.3 API Design Rules

- Use versioned routes and a documented OpenAPI contract.
- Use a stable response envelope with a safe error code, request ID and actionable message.
- Never return container names, internal paths, raw exception messages, private object keys or credentials.
- Apply ownership checks on every job endpoint; unguessable IDs are not a substitute for authorization.
- Use idempotency for job creation, upload completion, cancellation and deletion.
- Webhooks in a future API release must be signed, retry-bounded and protected against replay.

11. User Experience Requirements

Clear, adaptive and trust-aware

11.1 Required Screens and States

Screen / state	Required elements	Acceptance notes
Landing / upload	Product promise, supported format link, file picker, drop zone, limits and privacy summary.	No misleading “any file” claim.
File queue	File name, detected type, size, compatibility status, reorder/remove action.	Extension/signature mismatch is visible but does not expose internals.
Settings	Output page defaults and source-aware options only.	Do not display irrelevant settings.
Processing	Job stage, optional approximate progress, cancel action and safe help.	No fake percentage; keyboard/screen-reader announcements.
Success	Preview, download, delete now, expiry time and warning list.	Success only after output validation.
Failure	Specific user-safe error, next steps, retry when appropriate, support code.	Never show raw stack trace.
Format details/help	Accepted formats, limits, caveats, OCR notes and sample use cases.	Must match active format registry.
Admin console	Job metrics, safe failures, engine health, quotas, feature flags and retention checks.	Sensitive admin actions are audited.

11.2 Visual and Interaction Guidance

- Use one dominant upload action; secondary actions should not compete visually.
- Use plain labels: “Create PDF”, “Preview PDF”, “Delete files now”, “File type not supported”.
- Color must not be the only way to show a failed or completed status.
- Explain size and retention constraints before a user waits for an upload.
- Keep the batch list stable while individual files finish or fail.
- Preview should use a trusted PDF viewer and not execute active content from arbitrary files.

11.3 Product Copy Guardrails

Never claim

“Convert any file”, “perfectly preserves everything”, “100% secure”, “military-grade” or “fully GDPR compliant” without a precisely scoped and independently supported basis. Prefer “supports the formats listed here”, “formatting may vary for complex documents” and a concrete retention statement.

12. Error Handling and Observability

Make failures actionable and diagnosable

12.1 Error Catalogue

Error code	User message idea	System action
UNSUPPORTED_FORMAT	This file type is not supported yet. See supported formats.	Reject; no conversion worker.
TYPE_MISMATCH	The file extension does not match its contents. Check the file and try again.	Quarantine/reject; do not route by extension only.
FILE_CORRUPT	The file could not be read. Try opening and saving it again in its original app.	Fail deterministically; no repeated retries.
PASSWORD_PROTECTED	This file is protected and cannot be processed in this workflow.	Do not request or log passwords in MVP.
MALWARE_DETECTED	This upload was blocked because it failed a safety check.	Reject and record security event.
LIMIT_EXCEEDED	The file exceeds the current size or complexity limit.	Delete upload or expire promptly; show upgrade/simplify option.
TIMEOUT	Conversion took too long. Try a smaller file or simplify the document.	Terminate worker; clean scratch; bounded retry if transient.
ENGINE_UNAVAILABLE	Conversion is temporarily unavailable. Try again later.	Circuit-break, alert operations, avoid duplicate charge.
OUTPUT_INVALID	We could not validate the generated PDF. No file was marked ready.	Quarantine output, capture safe diagnostics, alert if rate spikes.
URL_BLOCKED	This address cannot be processed for safety reasons.	No fetch; log safe category only.
OCR_LOW_CONFIDENCE	The PDF was created, but some text may be inaccurate.	Deliver with warning and original-page preview.

12.2 Required Metrics

- Job count, success/failure rate, queue delay and conversion duration by format/engine version.
- Input/output size, page count, CPU time, peak memory and timeout rate by format family.
- Cleanup lag and number/age of expired artifacts; alert if deletion exceeds the published deadline.
- API latency/error rate, worker saturation, queue backlog and dead-letter count.
- Quota denials, abuse signals, malware detections and admin configuration changes.
- Costs per successful conversion and per format family, without retaining content.

12.3 Alerts and Runbooks

Define alerts for sustained failure-rate spikes, queue backlog, worker crashes, storage growth, retention cleanup delays, API availability, quota abuse and suspected cross-job access. Each alert must point to a runbook with first checks, containment steps, rollback option, owner and communication path.

13. Security, Privacy and Compliance Requirements

Treat every file and URL as untrusted input

13.1 Threat Model Summary

Threat	Example risk	Required mitigation
Parser exploit	Malformed DOCX/PDF/image crashes or compromises a worker.	Sandbox, non-root, seccomp/AppArmor or equivalent, no broad credentials, patching and malicious corpus tests.
ZIP/decompression bomb	Tiny archive expands to consume disk/memory.	Expanded-size, count, depth and ratio limits; streaming validation; timeout.
Path traversal	Archive member writes outside job scratch.	Normalize and validate paths; reject absolute paths, .. segments and symlinks as appropriate.
SSRF	URL renderer reaches cloud metadata or private service.	Connection-time destination validation, redirect checks, private-IP blocking, restricted egress and no ambient credentials. [R19]
Cross-tenant data leak	User guesses ID or reuses a download link.	Owner checks on every route, short-lived signed URL, private objects, random IDs and authorization tests.
Sensitive data persistence	Documents remain in temporary storage or logs.	Published TTL, cleanup worker + object lifecycle, delete action, log allowlist and periodic deletion tests.
Resource exhaustion	Huge pages, giant images, spreadsheets or OCR jobs.	File/page/pixel limits, per-engine memory/CPU/time caps and fair queue quotas.
Supply-chain/licensing	Engine dependency vulnerable or license incompatible.	SBOM, dependency scan, version pinning, license review and controlled upgrades.

13.2 Secure Upload Requirements

The upload design follows the broad defensive guidance in the OWASP File Upload Cheat Sheet: allowlisted types, signature/content checks, generated filenames, size limits, storage isolation, scanning where available and safe handling of untrusted files. [R18] The chosen control set must be reviewed against the actual worker stack and cloud platform, rather than treating this checklist as a complete security guarantee.

13.3 URL Rendering Requirements

URL-to-PDF is materially riskier than uploading a local file because the server makes outbound requests. Follow the OWASP SSRF guidance, including allow/deny network boundaries and defenses against redirects and DNS rebinding. Validate resolved addresses and ensure the HTTP client connects to the validated destination; do not trust a one-time hostname check. [R19]

13.4 Privacy-by-Design Controls

- Keep conversion content in a separate private artifact store from ordinary application records.
- Use a short TTL and double deletion mechanism: application cleanup plus object-storage lifecycle policy.
- Revoke read access at the control plane immediately on delete/expiry, even while physical storage cleanup completes.
- Keep only minimal, pseudonymous operational metadata for reliability and abuse prevention; define log retention and access roles.
- Publish what is stored, where it is processed, how long it is retained and which processors can access it.
- Complete a DPIA/privacy risk assessment if the intended use, jurisdictions or data types make one appropriate.

13.5 Compliance Planning

Before launch, obtain legal review for privacy notice, terms, copyright handling, payment rules and cross-border processing in each target market. This SRS is not a compliance certification. Target WCAG 2.2 AA and include manual accessibility testing. [R20]

14. Verification, Validation and Acceptance Criteria

Test with real files and adversarial inputs

14.1 Test Strategy

- Unit tests: format detection, option validation, quota reservations, lifecycle transitions and retention calculation.
- Integration tests: object storage, queue, worker dispatch, cancellation, retries, deletion and signed URLs.
- Golden-file tests: representative content/layout fixtures for every active format and engine version.
- Visual regression: render PDF pages to images and compare against approved reference outputs with format-specific tolerances.
- Security tests: malformed files, archive bombs, MIME spoofing, path traversal, SSRF, URL redirects, private IP and cross-tenant authorization.
- Load and soak tests: batch uploads, simultaneous jobs, OCR-heavy files, queue saturation, storage outage and worker crashes.
- Accessibility tests: keyboard-only paths, screen readers, visible focus, status announcements, contrast and responsive layout.
- Privacy tests: output expiry, failed-job cleanup, manual deletion, orphan cleanup, log inspection and restore procedure.

14.2 Acceptance Test Catalogue

Test ID	Scenario	Pass condition
AT-01	Valid DOCX with tables, headers, footer and mixed fonts converts into a readable PDF.	PDF opens; expected page count/layout checked; warnings displayed for known substitutions.
AT-02	XLSX with multiple sheets is submitted.	User can select sheets; PDF page scaling follows selection; printed output is readable.
AT-03	PPTX is submitted with a slide-range option.	Each selected slide appears in expected order; notes are excluded by default.
AT-04	Multiple JPG/PNG images are submitted.	Output page order matches UI order; rotation/orientation is correct; fit behavior matches selection.
AT-05	JSON or source code has a line longer than page width.	Output is readable and deterministic; long line wrap/overflow follows documented behavior.
AT-06	A file extension is renamed to a different type.	System identifies mismatch or rejects before conversion; no unsafe dispatch.
AT-07	A zero-byte, corrupted and password-protected file is submitted.	Each yields a specific safe error; no infinite retries.
AT-08	An archive contains ../ path entries and excessive expanded size.	Archive is rejected or unsafe entries excluded per policy; no writes escape scratch area.
AT-09	A URL resolves to localhost/private/metadata IP or redirects there.	URL job is blocked; worker cannot access the destination; security event logged safely.
AT-10	Two different users try to access each other's job/download.	Access is denied regardless of guessed job ID or URL tampering.
AT-11	A user deletes a completed job.	Download access is revoked, deletion queued and final artifact cleanup confirmed.
AT-12	A failed job reaches retention deadline.	Input, intermediate and output artifacts are removed within the published limit.
AT-13	Worker crashes mid-conversion.	Job reaches failed or retryable state; no false success; scratch artifacts cleaned.
AT-14	Malware scanner is unavailable for a high-risk format.	Configured fail-safe policy is followed; file is not silently processed.

Test ID	Scenario	Pass condition
AT-15	A screen reader user uploads and converts a file.	Controls, progress, errors and download actions are announced and operable.
AT-16	A batch contains a mix of supported and unsupported files.	Unsupported files are identified; supported files can continue if policy permits; order and individual outcomes are visible.
AT-17	Expired signed download URL is reused.	Access is denied; a new authorized link can only be issued while the job is still valid.
AT-18	Engine version is upgraded.	Full regression corpus is run; release is blocked if agreed quality/security thresholds fail.

14.3 Release Gates

- Every P1 requirement is implemented or formally deferred with owner and rationale.
- Every enabled format has a passing regression corpus, approved caveats, license status and a named engine version.
- No open critical/high security finding remains without an explicit, time-bound risk acceptance.
- Retention promise passes automated checks for successful, failed, cancelled and abandoned upload jobs.
- No cross-tenant access, public object storage exposure or secret/file-content logging issue exists in release tests.
- Load, accessibility, browser, rollback and incident-response checks pass for the beta scope.
- Product pages and plan limits match the actual support catalogue and runtime configuration.

15. Release Plan and Delivery Roadmap

Deliver reliable common formats before expanding the catalogue

15.1 Phased Plan

Phase	Scope	Exit criteria
Phase 0: Discovery / PoC	Validate 8-10 representative files per chosen format; test engine licenses; measure quality, speed, resource use and font behavior.	Approve MVP format list, worker base image, operating cost estimate and risk register.
Phase 1: Secure foundation	Upload, private storage, queue/state machine, worker sandbox, support registry, cleanup and logs.	Security review of upload and isolation; lifecycle deletion tests passing.
Phase 2: MVP converters	Office, images, text/markup, CSV/JSON/XML, source code, ZIP and OCR; URL route remains off until its security gate.	Golden corpus and acceptance tests pass for each enabled format.
Phase 3: Beta UX / reliability	Responsive UI, preview, batch, quotas, safe errors, metrics, admin engine flags and help content.	Usability, accessibility, load and incident-response reviews completed.
Phase 4: Public launch	Privacy/terms, support process, service limits, status page, billing decision, rollback and marketing claims review.	All launch gates signed by product, engineering, security and operations.
Phase 5: Expansion	API/webhooks, cloud imports, paid plan, more OCR languages, offline desktop mode, EPUB/email and specialist formats.	Each expansion has usage evidence and format-specific approval.

15.2 Suggested MVP Format Gate

Do not launch a format merely because a library can open one sample. For each P1 type, use at least a small golden set covering ordinary, complex, multilingual, malformed and large files. The initial number of fixtures should be based on risk; a practical baseline is 20-50 fixtures per high-volume format family and a smaller documented set for less common types, expanded as production failures reveal gaps.

15.3 Team Skills

- Product owner / UX designer to control scope and turn format limitations into understandable user choices.
- Backend engineer(s) for API, queues, storage, quota and lifecycle logic.
- Worker/platform engineer for container isolation, engine packaging and scaling.
- Frontend engineer for responsive upload queue, preview and accessible job states.
- QA engineer with document-rendering and visual-regression experience.
- Security/privacy reviewer and legal review for file processing, URL rendering, licenses and launch-market policy.

16. Risks, Dependencies and Open Decisions

Decisions that affect cost, quality and launch scope

16.1 Risk Register

Risk	Likelihood / impact	Mitigation / owner
False “universal converter” expectation	High / High	Product: publish explicit catalogue and exclude unsupported binary types.
Complex Office fidelity varies	High / High	Engineering/QA: reference corpus, native-output comparisons, visible warnings, optional commercial engine evaluation.
Untrusted parsers compromise workers	Medium / Critical	Security/Platform: sandbox, no credentials, constrained resources, patching and adversarial tests.
URL rendering permits SSRF	Medium / Critical	Security: feature flag off until connection-time enforcement and penetration testing pass.
Conversion cost exceeds freemium revenue	Medium / High	Business: measure CPU-time and storage cost; tier quotas; expensive features like OCR have separate limits.
Font licensing/substitution degrades layout	Medium / High	Engineering/Legal: approved font bundle, user warnings, licensing review and regression tests.
License obligations conflict with commercial model	Medium / High	Legal: dependency inventory/SBOM and license review before release.
Retention cleanup fails under outage	Low-Medium / Critical	Platform: dual cleanup, lifecycle policy, alerts, testable retention SLA and emergency cleanup runbook.
Specialist formats expand scope uncontrollably	High / Medium	Product: per-format admission gate and usage evidence before development.

16.2 Open Product Decisions

Decision	Recommendation for initial baseline	Must be decided by
Brand and domain	Keep AtlasPDF as a working name until trademark, social handle and domain checks are complete.	Founder / Legal
Launch region / cloud region	Choose based on initial customer base, processing latency, privacy rules and support capability.	Business / Platform / Legal
Guest vs account	Guest conversion for core usage; accounts only for quotas/history/billing and optional team features.	Product
Output retention	Proposed maximum 60 minutes after terminal state; make and test the promise before publishing.	Product / Privacy / Platform
URL capture in MVP	Keep disabled until SSRF review is complete; enable only with strict network policy.	Security / Product
Commercial engine	Begin with approved open-source candidate engines; evaluate paid engine only if quality and licensing economics justify it.	Engineering / Legal / Finance
Native Office fidelity	Define a “good enough” baseline by file family; do not claim equivalence to Office without evidence.	Product / QA
Pricing	Launch with configurable free-tier limits; set paid price after cost and competitor validation.	Business / Finance
Desktop offline app	Plan as a potential Phase 5 differentiator, not a blocker for the initial web MVP.	Product / Engineering

16.3 Dependencies

- Cloud storage lifecycle policy and queue semantics must be proven before the retention promise is marketed.
- Format and engine licenses must be compatible with the intended commercial distribution and deployment model.
- The format registry, API settings schema and frontend options must share a source of truth to prevent mismatches.
- Privacy policy, terms, abuse reporting and incident communications must be ready before public release.
- Operations must have alerts/runbooks and know how to disable a vulnerable engine quickly.

17. Success Metrics

Measure completion, quality, trust and cost

Metric	Definition	Target / initial operating rule
Conversion success rate	Successful validated jobs / jobs accepted for conversion, excluding pre-upload user abandons.	Baseline by format family; ≥98% on approved valid regression corpus. Production target refined by cohort.
Task completion	Users who obtain a valid PDF / users who start a conversion.	≥90% in moderated tests for common formats before wider launch.
Output quality defects	Reviewed cases with meaningful layout/content error / reviewed conversions.	Track by format/engine; no single aggregate should hide format-specific failures.
Time to PDF	Elapsed time from completed upload to ready output, by format, file size and pages.	Benchmark target: p95 ≤30 seconds for defined small common-office corpus under normal load.
Deletion SLA adherence	Artifacts removed within published TTL / artifacts due for deletion.	100% in automated test suite; alert on any production miss.
Error explainability	Failures mapped to a documented safe error code.	100% of terminal failures.
Abuse/security signal	Malware blocks, SSRF blocks, unauthorized access and policy violations.	Trend and investigate; zero confirmed cross-tenant access.
Cost per successful job	Worker, storage, queue and third-party cost / successful conversions.	Measured before plan pricing; OCR and URL capture tracked separately.
Support rate	Conversion support requests per 1,000 jobs.	Track by format and error code; use as prioritization input.
Retention cleanup lag	Time from expiry deadline to artifact removal.	Within the published deadline with a safety margin; alert before SLA breach.

Do not optimise only for the total number of supported formats. A format should stay enabled only while its reliability, output quality, security and operating cost remain acceptable.

Appendix A. Glossary

Term	Definition
Artifact	An uploaded file, intermediate file or generated output stored for a job.
Golden-file test	A repeatable test input with an approved expected output or visual/content reference.
OCR	Optical character recognition; extraction of machine-readable text from images or scans.
PDF/A	An archival subset of PDF intended to improve long-term reproducibility; it can change feature support and file size. [R21]
PWA	Progressive Web App: a web app that can offer install-like behavior in supported browsers.
SSRF	Server-side request forgery: an application is tricked into making unintended network requests.
TTL	Time to live: how long an artifact or access token remains valid.
Worker sandbox	A constrained execution environment for processing untrusted files.
Format registry	A versioned source of truth for support status, engine, options, limits and caveats for each format.

Appendix B. Research Sources

Official public product pages and technical guidance

Accessed/reviewed on 29 October 2026, except where the source page itself specifies a different publication/update date. These sources support market observations and engineering guidance; they do not constitute an exhaustive product audit.

[R01] Adobe Acrobat Help. Supported file formats for PDF conversion in Acrobat. https://helpx.adobe.com/au/acrobat/desktop/create-documents/create-pdfs/pdf-conversion-supported-file-formats.html

[R02] Adobe Acrobat Online. Online file converter, handling and common supported types. https://www.adobe.com/uk/acrobat/online/file-converter.html

[R03] Smallpdf. PDF Converter and public converter format scope. https://smallpdf.com/pdf-converter

[R04] Smallpdf. PDF tools catalog including OCR, PDF/A, HTML, EPUB, ZIP and Office formats. https://smallpdf.com/pdf-tools

[R05] iLovePDF. Pricing, plan limits and web/mobile/desktop availability. https://www.ilovepdf.com/pricing

[R06] CloudConvert. File Conversion API, format count, async jobs, options and storage integrations. https://cloudconvert.com/apis/file-conversion

[R07] CloudConvert. Privacy policy and data retention/deletion. https://cloudconvert.com/privacy

[R08] CloudConvert. Security and compliance practices, isolated containers and processing regions. https://cloudconvert.com/security

[R09] Convertio. Public converter homepage and format coverage claims. https://convertio.co/

[R10] Convertio. Privacy policy, file handling, processing and retention. https://convertio.co/privacy/

[R11] Zamzar. Public file converter and advertised format breadth. https://www.zamzar.com/

[R12] Zamzar Developers. Conversion API FAQ and representative supported format categories. https://developers.zamzar.com/faq

[R13] PDF24. PDF24 Creator: free offline Windows tool and local file handling. https://tools.pdf24.org/en/creator

[R14] PDF24. Webpage-to-PDF online conversion and temporary cloud processing. https://tools.pdf24.org/en/webpage-to-pdf

[R15] FreeConvert. Privacy policy and file retention/location. https://www.freeconvert.com/privacy

[R16] Microsoft Support. Save or convert to PDF/XPS in Office; output options, fonts and spreadsheet link caveat. https://support.microsoft.com/en-gb/office/collab-files/save-or-convert-to-pdf-or-xps-in-office-desktop-apps

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Appendix C. Release Readiness Checklist

Use as a sign-off record

Owner	Sign-off item	Done
Product	Public copy only describes actively supported formats and tested limitations.	☐
Product	Guest limits, batch behavior, retry semantics and billing/quotas are documented.	☐
Engineering	All P1 functional requirements are mapped to tests and release notes.	☐
Engineering	Every enabled engine is pinned, reproducible, monitored and independently disableable.	☐
Security	File upload and worker isolation threat model reviewed.	☐
Security	URL-to-PDF feature disabled unless SSRF acceptance tests and penetration review pass.	☐
Privacy	Retention statement is published only after end-to-end deletion tests pass.	☐
Privacy	Logs, metrics and error reports contain no file content, secrets or signed URLs.	☐
QA	Golden-file and visual regression corpus passes for every active format.	☐
QA	Malformed, large, archive-bomb, Unicode and unsupported inputs tested.	☐
Accessibility	WCAG 2.2 AA target test plan complete; keyboard and screen-reader checks passed for core flows.	☐
Operations	Alerts, on-call owner, rollback path, incident runbooks and service status communications prepared.	☐
Legal	Engine/font licenses, terms, privacy notice and launch-market obligations reviewed.	☐
Business	Cost per successful conversion and quota economics measured before final pricing.	☐